

Immediate actions on coal phaseout enable a just low-carbon transition in China's power sector

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HIGHLIGHTS

- A bottom-up job impact model is built to quantify the job impacts of decarbonization pathways of China's power sector.
- The net job impact of net-zero transition in China's power sector is positive.
- Early decarbonization and job training facilitates a socially equitable transition.
- Delayed mitigation action results in large mismatch of job creations and losses, hindering the ability of China to become carbon neutral by 2060.

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ABSTRACT

To meet the 2060 carbon neutrality target, China will need to phase out existing coal-fired power plants by 2050 or before. The electricity supply sector directly employs over 2 million and an additional 2 million when indirect employment including product and machinery production are included. To investigate the policy options and pathways available to meet the national climate goal while transitioning this jobs-intensive and economically powerful sector, we developed a plant-level job accounting model, and combine it with a power sector optimization model to assess job loss in the coal sector, as well as job creation in the burgeoning renewable power sector in China. We find that national and provincial policy actions to support an early and managed transition help to ensure a job-rich and both geographically and socioeconomically equitable shift from coal to clean energy. Specifically, the projected decline in fossil-fuel jobs can be fully offset by job new creation in the expanding renewable energy sector. Current COVID-19 economic stimulus plans include a potential new coal boom, where China could build up to 247 GW of additional plants, and thus delay the transition by a decade or more. We find that this action would result in up to 90% of coal-fired workers losing jobs between 2030 and 2040 without a clear pathway to absorb these workers in what will be an already mature clean energy economy. Provinces with massive coal fleets and limited renewable energy resources, notably Anhui, Henan, Hebei and Shandong, etc., would face a particularly disruptive mismatch of job gains and losses.

1. Introduction

Coal-fired power plants play a significant role in electricity

generation and thus also in CO₂ and SO₂ emissions in China today. In 2020, the installed capacity of coal-fired power plants in China totaled 1080 GW, accounting for 49% of national electricity capacity [1]. This

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fleet also comprises half of the world's installed coal capacity [2]. As a result, coal power sector is the largest contributor to China's carbon emissions, contributing around 40% of total national CO₂ emissions [3]. To meet the recently announced pledge of reaching carbon neutrality by 2060, China must decarbonize economy-wide, with an emphasis on phasing out the existing coal fleet by mid-century [4].

The replacement of coal with renewable energy is critical to China's climate plan, and in addition is critical to absorbing and re-employing workers in a sustainable energy economy. With over 4 million direct and indirect coal jobs today in China, planning the coal transition has become one of the key national policy debates that have profound impacts on global decarbonization. The existing literature has broadly explored the employment implications of this transition, with a number of studies focusing on the total number of jobs created by renewable expansion at the national level, especially the developed countries [5–7]. Most of these assessments find that there will be a net positive employment effect of coal transition, with more jobs created by the expansion of the renewable energy sector than those lost by the closure of coal capacity [8–12]. However, due to differing coal capacity and renewable energy resource endowments among regions, lay-offs from coal plants and employment opportunities in renewables will almost certainly distribute unevenly, leading to a spatial mismatch of job creations and losses [13–14]. This provincial (and thus, spatial) mismatch threatens to derail national decarbonization with very strong endorsement in some areas, and massive political push-back in others [15–18]. Moreover, as nature and skill requirements of solar and wind jobs differ largely from those of jobs in combustion-based thermal power plants [13,19–23], the transferability of employment between these sectors may be limited [14,24–26], leading to a potential structural misalignment in the job creation and lay-off.

The energy sector and employment aspects of China's decarbonization are challenging enough, given that this one of the largest sectoral transitions ever envisioned. Making this situation even more dire, however, are existing plans in China to potentially expand coal use. In the first half of 2020, the construction of an additional 40.8 GW coal-fired power capacity began across China. This total is more than the new coal installations for all of 2019 [27]. There are currently 88 GW coal-fired power capacity under construction and 159 GW approved in the pipeline waiting to be realized [28]. The current coal plant building spree in China is largely driven by a series of stimulus measures to promote economy and employment in response to the COVID pandemic [27,29–30]. The expansion of coal capacity will further lock China on a high-carbon path in the near term and thus make the needed emissions reduction trajectory far steeper if not impossible afterward [31–32]. Shutting down coal plants in an unmanaged and disorderly way may cause a huge number of coal workers laid off, and exacerbate the mismatch of job gains and losses, especially in hotspot areas with massive coal capacity. This “disorderly” transition then puts added pressure and unnecessary risks on the speed and extent of the job training investments and needed rate of business creation in the clean energy sector. Without a plan to generate new job opportunities at times and in provinces where coal phase-out will have the greatest impact, the Chinese government risks economic and political backlash that could cause the decarbonization to miss its 2060 target.

Three major research gaps of existing literature undermine their support for policymaking. First, most of the previous job impact assessments of China's low carbon transition have been conducted at the national level, overlooking the uneven distribution of lay-offs from coal plants and employment opportunities in renewables, and the resulting spatial mismatch of job creations and losses [13–14,24–25,33–35]. Second, some research has warned the consequence of delayed mitigation action, in terms of increasing mitigation costs, enlarging stranded assets, the accelerated requirement of renewable energies expansion [36–39]. Yet, to our best knowledge, no research has quantified the consequence of delayed mitigation action in terms of exacerbated labor inequality. [40] Third, most of the modeling studies on the low carbon

transition pathways of China's electric power sector have focused on NDC or 2 °C target [41–42]. Very few studies have assessed the cost-optimal technology evolution of the zero-carbon electricity system, in which fossil fuels may be phased out entirely [43].

To bridge these research gaps, this study develops a set of job and energy sector integration tools, conducts a first-of-its-kind assessment of the job impacts and potential misalignments for replacing coal-fired power plants with renewables, and proposes a roadmap to manage the jobs transition to align technological change, employment, and economic and political support for the decarbonization transition. Specifically, we first simulate the technological needs of immediate and delayed decarbonization pathways, and the consequential electric generating technology portfolios by applying SWITCH-China, a provincial power system optimization model. Then, we integrate these power-sector modeling results with a detailed bottom-up Job Impact Model for China Power System (JIMC) with a China-specific database to estimate the job losses in coal-fired power and job creations from renewable power under each decarbonization pathway. In addition to the total number of job changes at the national level, we assess the temporal, spatial, and structural disparities between job losses and job creations at the provincial level. This critical element of our methodology helps to identify hotspot areas with potential challenges of unemployment from coal phaseout, enabling advanced planning to manage compensation or re-training for laid-off workers.

2. Method, scenario and data

We present the modeling framework of this study in Fig. 1. By combining the SWITCH-China power sector model and the JIMC bottom-up job accounting model, this paper focuses on the job effects and misalignments beyond coal to renewable power in China's power sector under different decarbonization scenarios.

2.1. SWITCH-China model

SWITCH-China (<https://rael.berkeley.edu/project/switch/>) is a capacity expansion model used to explore the power system performance with minimized system cost under different policy scenarios (see Supplementary 1. The SWITCH-China model) [44–47]. SWITCH-China divides the geographic region of mainland China into 32 load areas, with Inner Mongolia disaggregated into East Inner Mongolia and West Inner Mongolia as they belong to two separate grids, with each of the other provinces representing one load area. The SWITCH-China model reflects the operating status of fossil fuel-fired generators at individual plant (unit) level, which then can be linked to detailed job account method. Different from integrated assessment models (IAMs), input-output models, or computable general equilibrium (CGE) models which are built based on aggregate approaches, SWITCH-China substantially improves the spatial granularity, permitting a comprehensive and detailed assessment of employment impacts and the socioeconomic implications.

We track the location, nameplate capacity, online year, operation status, combustion technology, heat efficiency, and fuel type of all the existing thermal electric generating units in China, by combining various databases and official statistics, including the databases of the Global Energy Monitor [28], the Platts World Electric Power Plant (WEPP) [48], the Carbon Monitoring for Action (CARMA) [49]. All the existing thermal generators are then incorporated into the SWITCH-China model with necessary aggregation. Each of the aggregated thermal power plant projects is given a binary decision variable in SWITCH-China to determine its operating status during each investment period [50]. Existing nuclear generators, totaling 49, are also modelled at unit level. Existing hydroelectric generators, including pumped hydro storage plants, operate subject to streamflow constraints. Intermittent wind and solar power stations are modelled based on detailed resource potential assessment and historical hourly load and generation profiles at provincial level.

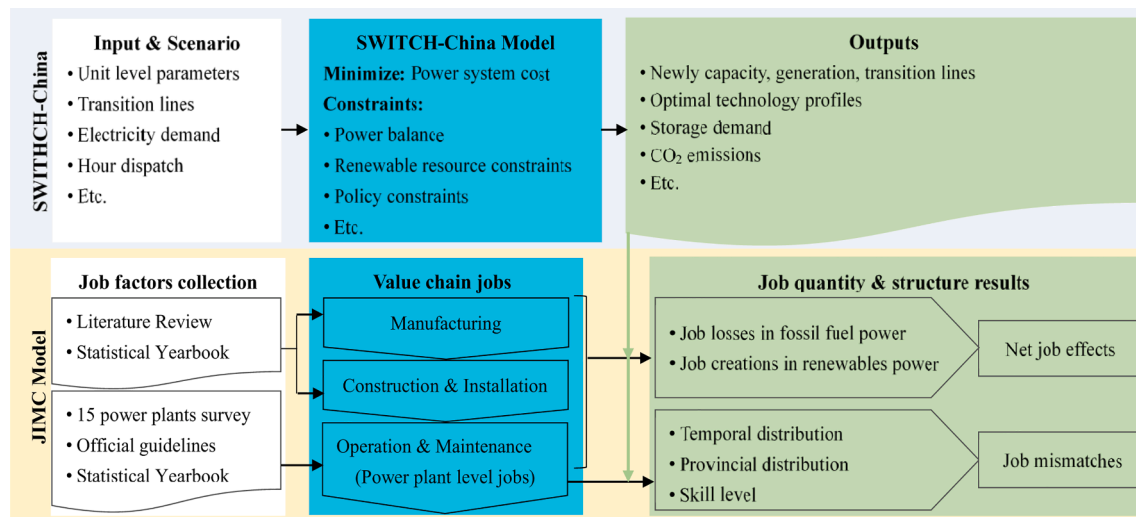


Fig. 1. Modeling framework adopted in the study integrating SWITCH-China and JIMC to examine the industrial, climate, and jobs impacts of different decarbonization pathways for China.

In addition to the operating units, there are also numerous coal power generators under construction or approved in China, which will likely be online in the near future. To reflect the potential coal boom witnessed in recent years, we collect information of coal plant projects which are under construction or permitted from Global Coal Plant Tracker database and recent editions of China Power Industry Annual Development Report, supplemented by news reports of new project announcements (see supplementary Fig. S2.1.) [28,51]. These coal power projects in the development cycle are assumed to be added into the coal fleet if ambitious mitigation actions are deferred (see Section 2.3 for details). For coal projects which are at the earlier phase of the pipeline (e.g. pre-permit development, announced), they are less likely to be eventually realized given the current policy trends in China, thus are excluded from this analysis. Various carbon-free energy sources and technologies, including utility-scale and distributed solar power, onshore and offshore wind, hydro power, nuclear, etc. are considered in SWITCH-China for future electricity mix. Multiple forms of storage, including pumped hydro, compressed air and battery storage are modelled to meet the hourly balance of electricity demand and supply. [46,52–53] Carbon capture and storage (CCS) and biomass combined with CCS (BECCS) technology are both important options to decarbonize the power systems in China. However, as the deployment of CCS and BECCS technology has continued to lag behind expectations and their social acceptance is relatively low, these technologies are excluded in future technology portfolios. Previous modeling simulations using SWITCH-China have demonstrated a relatively limited role of CCS and BECCS technology in a decarbonized Chinese electricity mix that mainly relies on wind and solar. Thus, excluding CCS and BECCS technology is not expected to alter the key findings in this study. Other long-duration storage technologies, such as molten salt, mechanical storage and hydrogen, are currently not included.

Different from the cost-optimal configuration in the SWITCH-China model, in the real world, China's electricity sector remains fundamentally government owned, operated, and regulated. The wholesale and retail electricity prices, feed-in tariffs, and transmission and distribution fees are set by the government. Despite some provincial pilots, dispatching of power plants is not in order of lowest marginal cost. The allocation of generation quota follows a “fair dispatch” principle, where generators in a given class, say coal-fired power plants, are allocated similar annual utilization hours by provincial governments. Meanwhile, most of the electricity generation assets and grid companies are owned by central or local governments. Investment or retirement decisions are not solely driven by economic incentives. However, China is now

undergoing an electricity market reform, aiming to liberalize the wholesale and retail electricity market, transition from fair to economic dispatch, and introduce more private investment in generation and wholesale [54]. Once big steps towards deregulated market are taken, the electricity market structure in China will then be closer to the cost-optimal configuration of the SWITCH-China model.

2.2. Employment impact evaluation method

According to the previous studies, input–output and analytical models are two main methods to evaluate the employment impacts [7,55–57]. Compared with the input–output model, the analytical method is more flexible, thus is found to be more suitable for granular employment effects studies [5,9,58]. Further, it is easy for the analytical model to assess the future changes under different timespans while it is very difficult for the input–output model to do so [6,11,58]. As this study aims to address the quantitative changes, distributional effects and potential misalignments when fossil fuel workers switch to renewable jobs, the analytical method is adopted.

Here we utilize our bottom-up job analytical model-Job Impact Model for China Power System (JIMC) to evaluate the job changes and distributional effects. Based on the value chains of different electricity technologies, the JIMC model tracks the interlocking jobs in the equipment manufacturing, construction and installation (C&I), and operation and maintenance (O&M) of multiple electricity technologies, including coal, natural gas, hydro, nuclear, solar, wind and storage. Equipment manufacturing jobs refer to those offered by electrical equipment industries, like the manufacturing of boilers, turbines, modules, etc. Construction and installation jobs involve the phase of building a new power plant. Operation and maintenance employees are hired to ensure the everyday routine operation of the power plants. As shown in Fig. 2, operation and maintenance jobs are defined as direct employment in this study, as they are directly connected to the electricity production. Construction, installation and manufacturing (CIM) are secondary activities, which provide goods or services to build a power plant, not directly produce the electricity. Thus, employment in these phases is classified as contributing to indirect jobs. Induced employment, which associated with changes in expenditure, is not included in this paper.

Jobs of different value chains vary significantly in both time and location dimensions. Once the CIM phase is finished (which typically lasts 2–3 years for coal power plants, 10 years for nuclear power plants, 1–2 years for wind and solar stations) [59], the O&M jobs remain

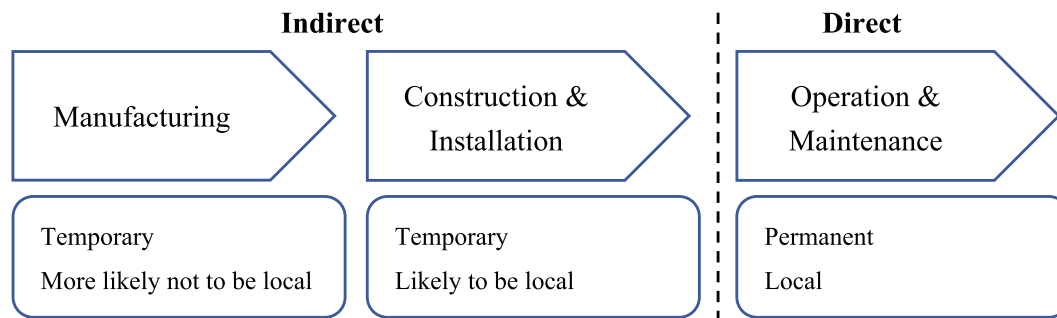


Fig. 2. The distribution of different job categories tracked in JIMC along power sector value chains quantified in this study.

permanently through the entire lifetime of a power plant [7,57]. Geographically, manufacturing jobs do not necessarily locate in the same place as the power plant, while O&M employees do [24]. In this regard, we measure employment coefficients in the equipment manufacturing, construction, and installation stages as job-years per megawatt over the project duration. One job-year (also known as “full-time equivalent job”) refers to full-time job for one person during the period of one year [7]. O&M jobs, regarded as permanent on-site jobs existing in the entire lifetime of the power plants, are measured as jobs per megawatt [55]. To combine the two groups of jobs, we convert the job-years per megawatt to jobs per megawatt by averaging them over the service lifetime of the power plant [7,60].

As described in Fig. 1, jobs are quantified by multiplying electricity generation capacity and corresponding value chain employment factors. To achieve a high-resolution job assessment, power plant level job factors are required in addition to electricity mix derived from SWITCH-China. Many studies have estimated the employment factors of power technologies, and Wei et al. (2010) [7], Cameron et al. (2015) [56], Ortega et al (2015) [61], and Mu et al. (2018) [9] conducted detailed reviews. However, most of the extant literature focus on developed countries [9,23,62]. Due to different levels of automation and labor productivity, employment factors may vary significantly in China [9,26]. Moreover, current literature only reports the average employment factors of coal plants, overlooking the fact that job coefficients differ substantially in small and large coal power plants. We therefore obtain employment factors of the O&M phase for coal, natural gas, wind, solar and hydro plants from official workforce guidelines in this study [63–66]. Workforce guidelines set standards for the numbers of jobs of each occupation (e.g. manager, technician, operator, etc. ...) required to operate a typical power plant of different installation size (Table 1). For jobs at the CIM phase, as there are no open-access national data sets or guidelines, so instead we utilize median estimations from extant literature are used for each of the power generation technologies (Fig. 3, Supplementary Table S2.2–2.3.). The adoption of the China-specific job factors enables a more detailed plant-by-plant estimation especially for the existing coal fleet. Following the methodology, we could derive the number of direct and indirect involved in: 1) the construction, installation, manufacture phase of each additional technology and 2) the ongoing operation and maintenance phase of the existing units.

In addition to the total number of jobs, we also pay close attention to the skills required and regional distribution of these jobs. According to the Ministry of Human Resources and Social Security (MOHRSS) of China, the professional qualification levels are classified into five categories from highest to lowest: senior engineer (Level 1), engineer (level 2), senior technician (level 3), medium technician (level 4), junior technician (level 5) [67]. We conducted a survey of 7 coal-fired power plants and 8 renewable power plants from 11 cities in 5 provinces in China (see Supplementary Table S2.4.), supplemented by another survey done by Institute for Labor Studies and Ministry of Human Resources and Social Security [68], deriving the skill level of coal, wind and solar power plant employment. Assuming the existing skill remains

Table 1
Jobs per MW in O&M stages by electricity technology.

Electricity technology	Size: MW	O&M factors: jobs/MW
Coal [63]	<100	5.18
	100–300	2.70
	300–600	1.21
	600–1000	0.65
	≥1000	0.28
Natural Gas [63]	<100	5.18
	100–300	2.70
	300–600	1.21
	600–1000	0.65
	≥1000	0.28
Wind [64]	<200	1.55
	200–400	0.43
	>400	0.34
Solar [65]	≤50	2.00
	50–100	0.81
	100–200	0.44
	>200	0.19
Hydro [66]	<15	8.16
	15–35	5.08
	35–100	2.64
	>100	1.01
Nuclear*	—	0.43
Storage*	—	0.4

Note: * comes from literature review, see supplementary S2.2–2.3 for more detail.

unchanged, we can project the number of jobs of different skill levels associated with future electricity mix.

2.3. Scenario setting

There are various transition pathways to meet the stringent sectoral carbon budget with differing temporal distribution of mitigation ambitions [4,36,39]. We designed two stylized transition pathways to achieve a fully decarbonized power sector in China (Table 2). We used SWITCH-China to investigate the future evolution of electricity technologies under these pathways. As a heatedly debated topic recently in China, when to stop building new coal projects, and when and how to phase out the existing coal fleet are the main policy components we address in the scenario design.

Under the Orderly net-zero 2050 (hereinafter referred to as NZ2050_O) scenario, an early and orderly transition path is envisaged with CO₂ emissions declining to net-zero by 2050, a target that is in line with China’s 2060 carbon neutrality pledge and could limit global warming to below 1.5 °C. A prohibition of new coal plant build will be implemented immediately, canceling all the planned coal projects currently in the pipeline. Then a linear decrease from 2020 until 2050 where emissions reach zero is enforced. As near-term decision making has already incorporated long-term climate constraints, the transition to a carbon-neutral power system under the NZ2050_O scenario occurs in a

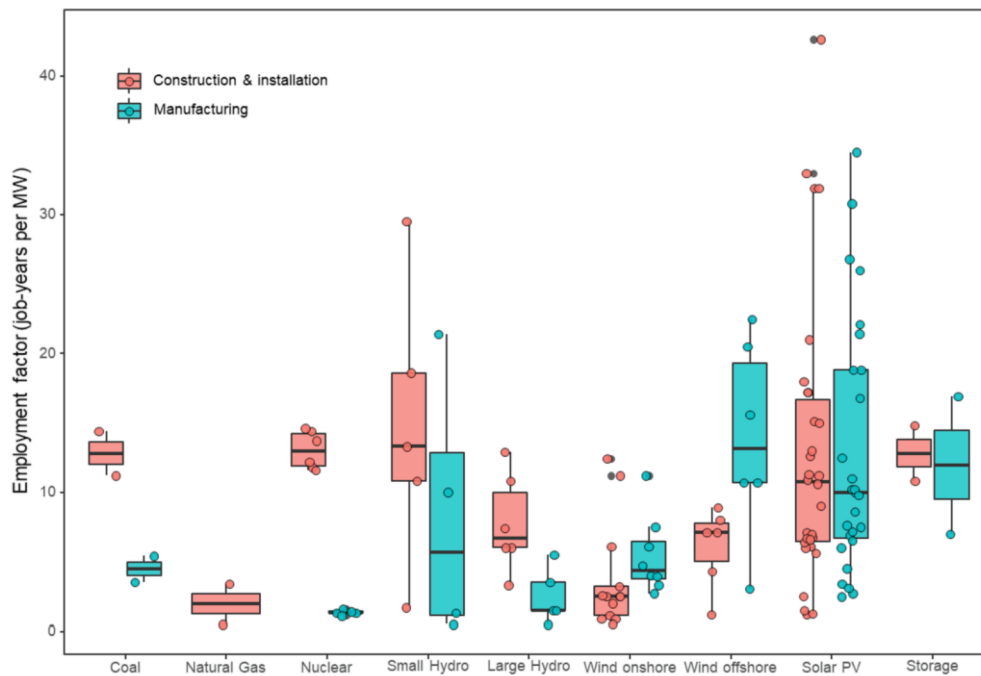


Fig. 3. Job-years per MW in CIM stages by electricity technology.

Table 2

Scenario definition.

Scenario	Abbreviation	Scenario description
Reference	REF	Keep adding new coal-fired power plant; No explicit climate targets after 2020;
Orderly net-zero 2050	NZ2050_O	No newly installed coal generation after 2020; Early retirement of coal-fired power plants after 2020; Emissions decrease linearly to zero by 2050;
Disorderly net-zero 2050	NZ2050_D	Keep building new coal-fired power plants until 2030; All the coal units operate until the end of lifetime without early retirement before 2030; Emissions decline rapidly after 2030 to meet the same carbon budget during 2020–2050 as the NZ2050_O scenario;

gradual, anticipated and managed way.

In contrast, the Disorderly net-zero 2050 (hereinafter referred to as NZ2050_D) scenario represents an alternative path in which ambitious mitigation actions are delayed for another ten years, partially driven by fossil fuel-reliant post-COVID economic stimulus plans. Under this alternative scenario, no strict restriction on new coal build is enacted before 2030, and all the coal projects that are under construction or approved by the government are assumed to be added into the pipeline eventually. Total installed capacity of coal-fired power plants will reach 1200 and 1300 GW by 2025 and 2030 respectively, in line with the outlook proposed by China Electricity Council (CEC) [69] and China Electric Power Planning & Engineering Institute (EPPEI) [70]. The low initial mitigation ambition leads to quick depletion of the carbon budget, and accelerated CO₂ reduction measures are required afterwards to meet the same sectoral carbon budget during 2020–2050 as the immediate action scenario. This opposite transition pathway may involve sudden, disruptive and disorderly changes in the power sector.

In addition, we include a hypothetical reference scenario (REF) wherein no explicit carbon policies are implemented after 2020. Although highly unlikely, the counterfactual REF scenario is intended to serve as a baseline for understanding additional policy impact in power generation technology evolution and job losses and creations.

3. Results

3.1. Low-carbon transition pathways of the power sector in China

In Fig. 4 we compare the future evolution of technology mix over 2020–2050 under different scenarios. Driven by stringent climate targets, the NZ2050_O scenario, compared with the REF scenario, leads to a more rapid phase-out of coal capacity, from over 1043GW in 2020 to 823GW in 2030, and to zero eventually by 2050. The pace of coal retirement keeps steady at around 36GW annually, enabling a ‘soft landing’ for the coal sector. In the meantime, wind and solar capacity would grow faster to 2122GW and 3327GW respectively in 2050, replacing coal as the dominant electricity source. The need for grid stability would result in increasing storage capacity to support the integration of variable renewable energies. Assuming all the planned coal project pipeline being eventually realized, the NZ2050_D scenario predicts a continuous growth of coal capacity before 2030, thus the overwhelming dominance of coal in China’s power sector remains. Accordingly, renewable energy such as wind and solar would be installed at a much lower speed compared with the early action scenario during the period. Delayed mitigation actions in the near term would then lead to much more radical changes in the electricity structure afterwards compared with NZ2050_O scenario in which ambitious actions are taken right after 2020. Massive early retirement of coal-fired power plants would inevitably occur after 2030 to reach net-zero emissions by 2050. Wind and solar soar between 2030 and 2050 to meet the growing electricity demand. Meanwhile, the capacity of storage expands to 911GW by 2050 to stabilize the intermittency caused by large-scale utilization of wind and solar energy. Total installed capacity of hydro and nuclear power plants remains relatively constant during the period across all three scenarios, as exploitable hydropower resources have been largely utilized, as well as a strict limitation of nuclear projects within coastal provinces due to security concerns. We note, however, that re-purposing coal company human resources and infrastructure is possible to engage in geothermal power, the Hydrogen economy, and other areas.

Comparing with the orderly transition pathway, 220 GW more coal-fired capacity would be invested in under the disorderly pathway by 2030. This amounts to a huge sunk-cost that is both under-utilized, and

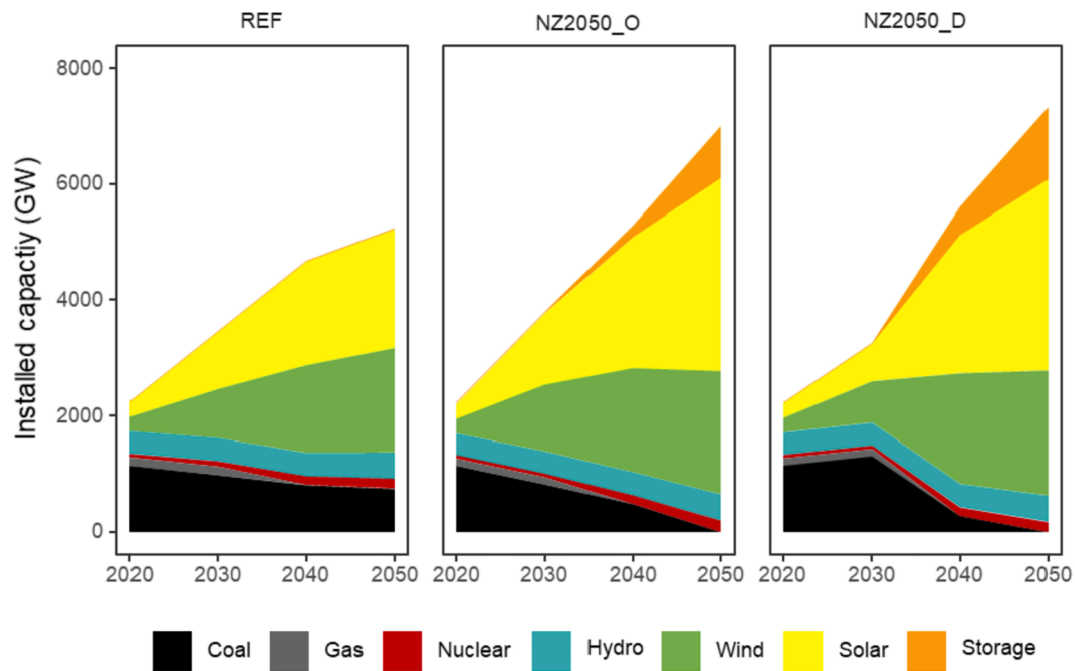


Fig. 4. Power sector decarbonization pathways in SWITCH-China.

retards the development of job retraining and other efforts to build the work-force China will need in a decarbonized future.

We can therefore robustly conclude that it would be a significant waste of investment and exacerbate the overcapacity of coal power, as all the newly installed coal plants need to be closed prematurely later. Consequently, the NZ2050_D scenario foresees a much more sudden and harder landing for the coal sector afterwards. Meanwhile, the required installation rate of wind and solar reaches 294 GW annually from 2030 to 2040, almost three times as high as the historical maxima, making this late and disorderly transition highly challenging and risky.

3.2. Power sector employment and their changes under three scenarios

Based on our job model results, 4,589,000 workers are hired by the power sector along the supply chain in 2020. Among them, 2,099,000 people are working in power plants directly, with 44% in thermal power plants, 18% in hydro and nuclear plants, and 37% in wind and solar power plants. As renewables are usually more labor-intensive than coal power, replacing fossil fuel-oriented energy with renewable energy can generally bring a positive employment effect. As shown in Fig. 5, Under the REF scenario, which remains a coal-heavy power structure with modest renewable expansion, the total number of jobs increases to 5,747,000 and 7,862,000 by 2030 and 2050, respectively. Under the two policy scenarios, though the fossil fuel-dependent jobs drop

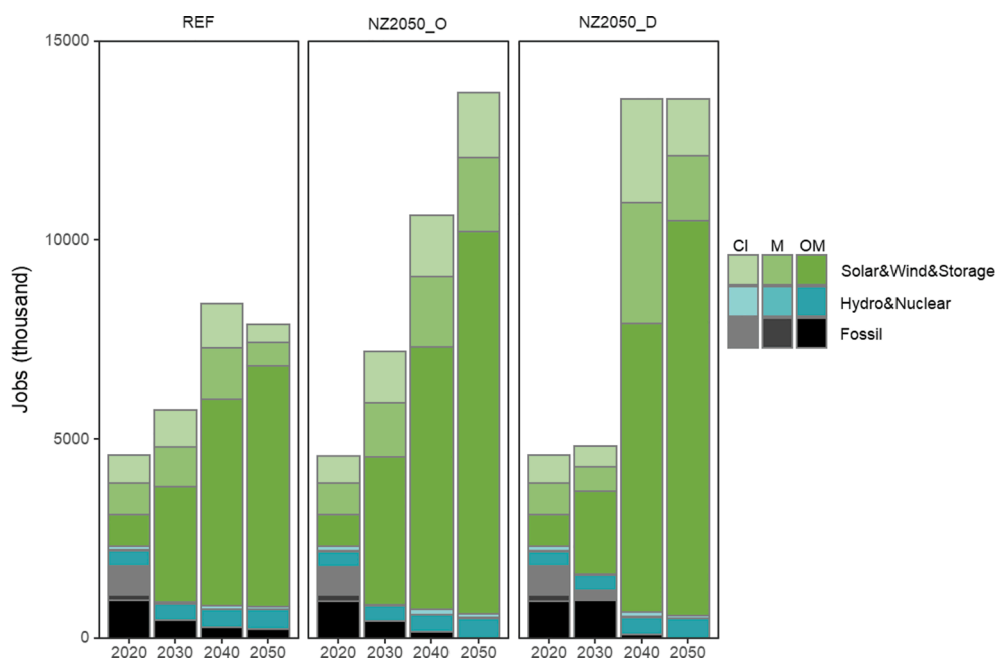


Fig. 5. Power sector value-chain jobs under three scenarios.

dramatically, the total job opportunities keep growing and becoming larger than in the REF scenario. Under the orderly transition pathway, total job opportunities add up to 7,182,000 in 2030 and 13,701,000 in 2050, which are 25% and 74% larger than the REF scenario. The disorderly transition pathway brings similar additional job opportunities in total than the baseline case by 2050. Employment in hydro and nuclear power plants remains steady as a result of the stagnant hydro and nuclear capacity across scenarios.

Along the value chain, indirect jobs in the CIM phase account for around 54% of total power sector employment in the base year. CIM jobs for the coal sector diminish rapidly after 2030 in the NZ2050_O scenario as new coal build will stop immediately, and CIM employment for the renewable energy sector keep steady at around 2,600,000 with the gradual expansion of wind and solar capacity. In contrast, CIM jobs for the coal sector remain as planned coal projects continue to be completed before 2030. Along with the unprecedented expansion of renewable energy from 2030 to 2040 in the NZ2050_D scenario, 3,072,000 jobs would be created to manufacture, construct and install these wind turbines and solar panels. The proportion of indirect CIM jobs declines gradually regardless of the scenario variants, because when the stock of power units accumulates, the pace of new capacity installation slows down.

Although both the orderly and disorderly transitions create a similar scale of employment opportunities by mid-century, there are major differences in their temporal pattern. Under the NZ2050_O scenario with immediate mitigation actions, the build of new coal capacity would stop immediately, and more labor-intensive renewable energies would replace coal capacity right after 2020, leading to substantial employment growth between 2020 and 2030. In contrast, a new coal boom delays the transition steps, results in a demand for 111,000 additional coal workers in the near term. Total jobs in the power sector keep stable

before 2030 in the NZ2050_D scenario, as a result of the continuous completion of coal projects and delayed development of wind and solar. Along with the unprecedented retirement of fossil fuel plants and deployment of wind and solar energy between 2030 and 2040 to meet the eventual net-zero emissions target by 2050 in the power sector, workers in the coal plants are laid off almost entirely, while total employment in renewable energy is nearly triple within a decade.

3.3. Direct job losses and creation: sectoral focus and spatial distribution

Although the jobs created by renewables outnumber the job losses from fossil fuel at the national level, it does not necessarily lead to an equitable transition as greater employment mobility brings a high risk of misalignments both temporally and spatially. This section presents the potential temporal, spatial, and skill level distribution of direct jobs and possible mismatches in orderly and disorderly scenarios. Here we only focus on direct O&M jobs for two reasons: 1) direct O&M jobs are permanent employment while CIM jobs only last during the manufacture, construction and installation phases, 2) O&M workers tend to strongly attach to the local community and have a lower possibility to migrate when losing jobs.

In terms of the quantity, job creations in renewable power under two different transition pathways are similar by mid-century, and both outnumber the job losses caused by the retirement of coal plants (Fig. 6). However, they can yield starkly different temporal distributions of job displacement and levels of risk exposed to laid-off workers. In the NZ2050_O scenario, as near-term decision making has already incorporated long-term climate constraint, coal capacity is decommissioned in a smooth and managed way during the period from 2020 to 2050. The fossil fuel employment nearly halves every decade, and the jobs created by expanding renewable capacity increase at a steady pace of 2.4–2.9

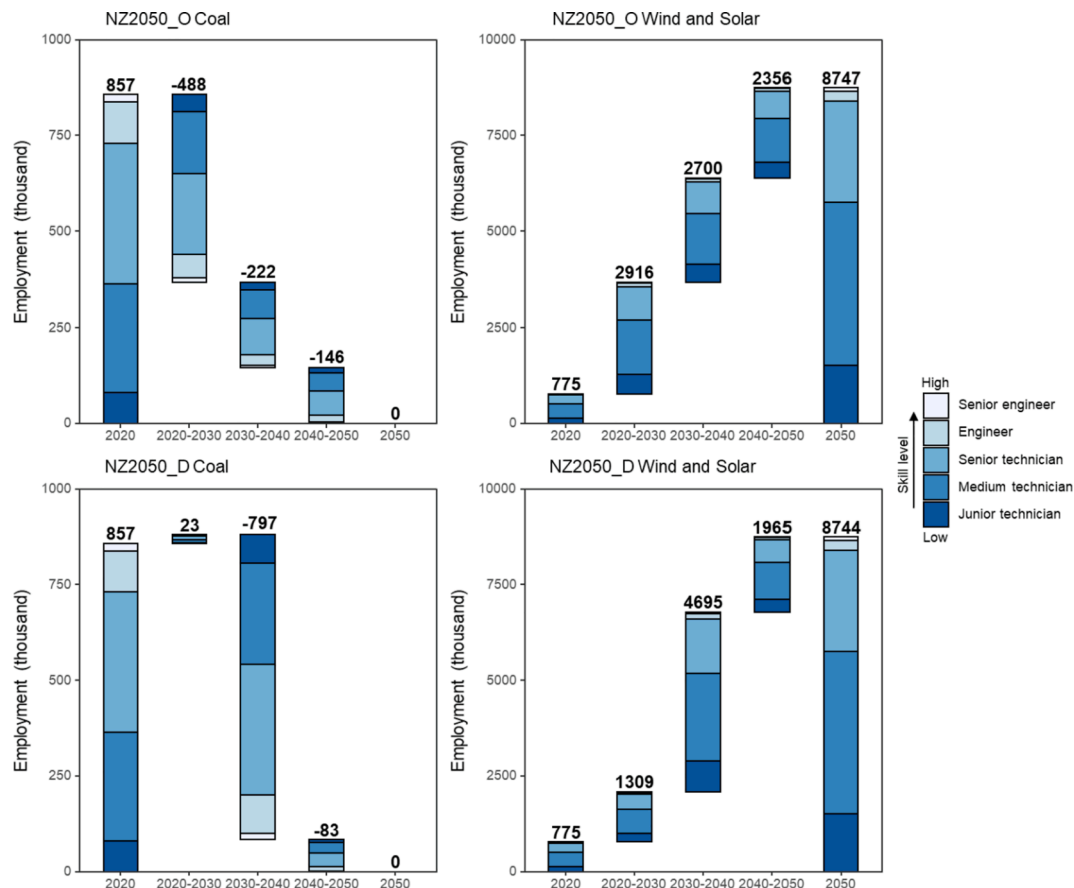


Fig. 6. Temporal distribution of job losses and job creations and their skill level breakdown.

million every ten years. The gradual evolution in power technology structure and the managed job displacement and creations could provide the government and society enough time to retrain, relocate or compensate affected workers.

In stark contrast to the early and steady transition pathway, there is a significantly uneven distribution of job losses and creations among decades in the late and rapid path. Before 2030, only a few coal units retire when reaching their lifetimes, most of which are small and inefficient. Due to the completion of the coal project in the pipeline, an additional 23,000 workers are hired in coal-fired power plants. Consequently, the deployment of renewable energy slows down, creating less than half of the employment compared to the NZ2050_O scenario during 2020 to 2030. A sharp reduction in coal capacity afterwards leads to massive displacement of coal workers. This sudden and unpredictable coal phase-out would expose 797,000 workers – over 90% of total employment in the coal power sector – to the risk of losing jobs between 2030 and 2040. Meanwhile, wind and solar plants need to be built at an unprecedented rate after the initial slowing down, generating almost twice as large requirements in new labor force from wind and solar power compared with NZ2050_O scenario. The sudden increments in labor force demand from wind and solar would also cause a possible supply shortage in the labor market. To mitigate the potential labor supply bottleneck, substantial coordination between government agencies, universities and colleges, occupational-skill training institutes, and firms are required to meet the evolving need of both workers and employers.

A second concern is that there is also an uneven skill level breakdown of the employment in coal-fired power plants and wind and solar stations. Based on our survey, among coal power employment, 15% are regarded as high skill workers (senior engineer or engineer), 76% are medium skill (senior technician, medium technician), and around 9% are low skill workers (junior technician). In comparison, the proportion

of high, medium and low skill workers in wind and solar power employment is 4%, 79% and 17% respectively. Assuming the skill structure of labor force in coal and renewable power plants remains unchanged, we can project the scale of unemployment and demand for labor force of varying skill levels under different scenarios in the future. Compared with the smooth and orderly transition pathway, lay-off of coal workers and demand for workers in wind and solar stations are much more concentrated temporally. During 2030 and 2040, over 74,500 workers with limited skills are laid off due to the closure of coal plants in the NZ2050_D, over three times larger than the unemployment in the NZ2050_O scenario during the same period. Due to their relatively low education and skill level, it would be difficult to retrain and relocate these labor forces. This huge unemployment wave within a short period will be a big challenge to the government and the whole society. Correspondingly, wind and solar power stations need to hire 188,000 and 3,690,000 new workers with high and medium skill levels respectively within ten years. The explosive growth in demand for renewable plant employees also puts great pressure on the supply of skilled workers in the labor market. Therefore, this “disorderly” transition puts added pressure and unnecessary risks on the speed and extent of the job training investments and needed rate of business creation in the clean energy sector.

The projected job displacement and creation and the associated challenges in both dealing with unemployment and meeting demand for labor supply may be exacerbated at the provincial level as they may be disproportionately distributed among regions. The top ten provinces with the largest existing coal fleet account for 65% of national installed capacity in the base year. Accordingly, risk of unemployment from coal plants closure would also be concentrated in these provinces. Therefore, we take a close look at the ten hotspot provinces with the largest coal capacity and workers.

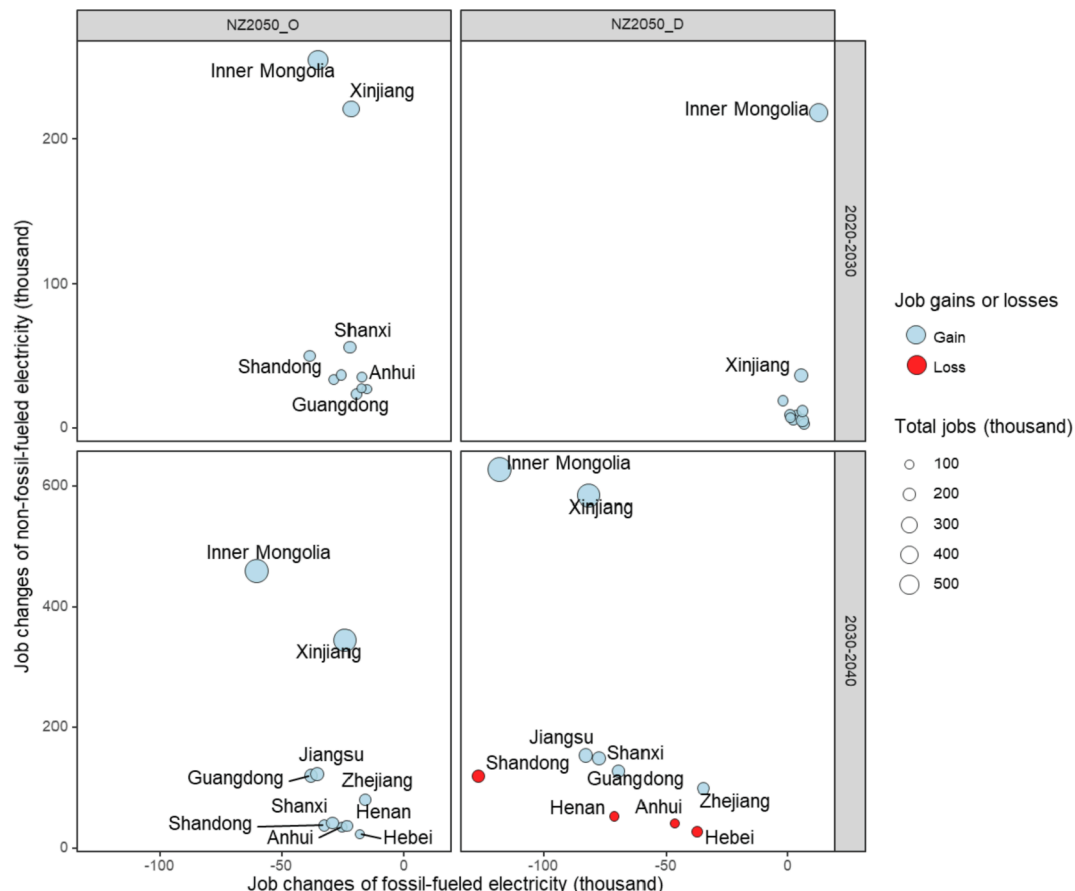


Fig. 7. Spatial mismatch of job losses and creations in top 10 coal rich provinces.

As shown in Fig. 7, job disruptions of coal power sector, as well as the employment opportunities of renewable energy foreseen in the orderly and disorderly transition pathways, are unevenly distributed across these provinces. In the NZ2050_O scenario, as ambitious actions are taken right after 2020, coal plants are closed at a steady speed across decades. Inner Mongolia and Xinjiang, two provincial regions with rich renewable energy resources, are forecast to install the largest amount of new wind and solar capacity, and hence be the sites of the largest job demand. From the perspective of the total number of job losses and creations, more job opportunities are supplied by newly installed renewable capacity than unemployed workers from decommissioned coal power plants, leading to a net positive employment impact in all the ten provinces during the entire period. Although clean energy job creations surpassing fossil-fuel job losses does not necessarily mean that the low-carbon sector could employ all the displaced fossil fuel workers, it does reduce the risk of large-scale unemployment.

In the NZ2050_D scenario, most of the existing fossil fuel plants will keep operating before 2030, thus fossil fuel workers can retain their jobs. However, a radical change in power structure would follow this temporary prosperity of coal. Fast-paced layoffs would occur across the country, reducing jobs in coal-fired power plants by nearly 90% within a decade. Shandong and Inner Mongolia, among the provinces with the largest coal employment, are the hardest hit areas. In provinces where resource availability is limited, development of renewable energy may not be able to offset the unemployment from coal in terms of absolute numbers, leading to net job losses in the power sector. In this case, four provinces, i.e., Anhui, Henan, Hebei and Shandong, are projected to suffer from net job losses, with a total deficit of over 40,000 workers. Therefore, the sudden closure of coal plants will certainly exacerbate job market inequalities.

4. Discussion and conclusion

4.1. Main findings and policy implications

By combining a bottom-up power sector optimization model and a detailed job assessment model, this study quantified the job losses and creations and their distributions on timeframe, power generation technology, geographic location, project phase and skill levels under different power sector decarbonization pathways in China on a regional basis from 2020 to 2050. The results demonstrate that decarbonizing the power system can create 6.5 million more job opportunities by 2050 in general, but different transition pathways yield starkly different results. An early and steady path entails a smoother evolution of power structure changes which could better accommodate the employment and social aspects of the transition. This gradual and managed evolution could facilitate a soft landing of coal fleet, reduce the risk of large-scale unemployment, and deliver a technically feasible and socially equitable transition. In contrast, realizing planned coal projects currently in the pipeline would delay the transition steps, resulting in a sudden and unpredictable coal phase-out afterwards, exposing over 90% of coal plant employment to the risk of job displacement during 2030 and 2040. This sudden and disorderly transition would exacerbate labor market inequalities, as job losses are concentrated within a shorter time period and in some hotspot regions.

Power sector decarbonization can contribute to more jobs in the long run as renewables will create more jobs than the job elimination in the coal-fired power plants. However, workers who lost their job during the decarbonization transition are not necessarily those who will get renewable jobs. To make the coal transition just and affordable, we should pay more attention to the job losses, and the potential misalignments between job losses and creations, including their structure and distributions. Given the large fleet of coal-fired power plants in China, phasing out coal capacity of this magnitude not only displace individual workers, but also disrupt local communities and decay local economy. China has already made a series of policies to retrain, relocate

and compensate laid-off workers when eliminating excess production capacity in heavy industries since 2016 [71–74]. However, a comprehensive policy framework to ensure a just transition for fossil fuel workers in China is still nascent.

To alleviate the negative impacts of the energy transition, holistic and proactive planning, workforce retraining programs, job placement assistance, and other supportive policies are vital to facilitate laid-off coal workers to build skills and transfer to other industries. What is happening in coal power sector would also be a harbinger for other carbon-intensive industries as the deep de-carbonization transition advances. Moreover, given the magnitude of future labor demand from expanding renewable capacity, large-scale and sustained support of workforce supply, and substantial coordination between government agencies, universities and colleges, occupational-skill training institutes, and firms are required to mitigate potential labor supply bottleneck.

4.2. Limitations and future work

There are a number of limitations in this study. First, this research quantified value chain employment of the power sector, including equipment manufacturing, construction and installation, and operation and maintenance phase, without considering upstream fuel supply industries, especially the coal mining industry. Coal mining industry hires over 2 million workers in China in 2020, most of which have low education and skill level, and are concentrated in coal abundant provinces. Therefore, it is more difficult to help coal miners transfer to other industries once they are displaced from the current jobs. This paper mainly focuses on the on-site job losses and creations of coal plant closure and renewable plant development. Quantifying the coal mining job losses from coal-fired power plant closure requires detailed mapping of trade and transport networks between individual coal mines to power plants, which warrants further modeling work. Although beyond the scope of this paper, a detailed quantitative assessment of potential job losses of the coal mining industry is critical to ensure a just low-carbon transition.

Retrofitting of existing coal power plants with CCS and BECCS is not included in the model as a carbon mitigation technology option because of its low technology maturity, high cost and low social acceptance. The adoption of CCS helps to extend the lifetime of coal plants under strict climate policies, and retain jobs that would be otherwise lost due to retirement of coal capacity. Modeling studies with perfect foresight over the investment horizon have demonstrated that a decarbonized Chinese electricity mix would mainly rely on wind and solar in the cost-optimal pathway, avoiding large-scale utilization of CCS and BECCS technology [43–44]. Thus, the aforementioned limitation might slightly impact the results but is not expected to modify the main findings obtained in this study.

Third, this study adopted constant job factors without considering possible changes in the future. As a result of economy of scale and labor productivity improvement, the employment multipliers of renewable electricity are likely to decrease in the long term. Particularly, those operation and maintenance jobs associated with the routine operation of wind turbines and solar panels may be exposed to displacement arising from greater automation. However, employment factors in the coal power plants are likely to remain unchanged in the future given the maturity of coal-fired power technology in China. Therefore, considering the possible decrease of employment factors in renewable power may result in more severe job mismatches.

Fourth, as this paper presented a partial equilibrium model of power sector alone, we considered the labor market as given, without considering the linkage between labor supply cost and power system evolution. Envisaging the challenges of highly skilled labor supply facing renewable expansion, the Chinese government has formulated and implemented a set of proactive employment policies, including reshaping the national skills development system in order to reduce the skills mismatch, promote vocational education and training, and mobilize different ways of learning to cope with the need for highly skilled

workers [75–76] If implemented effectively, these measures would help to alleviate the labor supply shortage caused by renewable expansion. Labor cost, which has already been considered in the cost metrics of the SWITCH-China model, would not be expected to rise significantly, thus would not alter the modeling results substantially.

CRediT authorship contribution statement

Xiaoli Zhang: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Visualization, Writing – review & editing. **Xueqin Cui:** Conceptualization, Software, Writing – review & editing, Visualization. **Bo Li:** Software, Writing – review & editing. **Patricia Hidalgo-Gonzalez:** Software, Writing – review & editing. **Daniel M Kammen:** Project administration, Funding acquisition, Supervision, Writing – review & editing. **Ji Zou:** Resources. **Ke Wang:** Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apenergy.2021.118401>.

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